

Question

Deduce and write the equations for the following chemical reactions (requiring the coefficient ratio to be the simplest integer ratio).

1-1 Phosphorus trifluoride decomposes upon heating, producing a molecule with a tetrahedral structure.

1-2 Manganese(II) nitrate hexahydrate reacts with acetic anhydride to produce manganese(III) acetate and release a reddish-brown mixed gas.

1-3 Iodine phosphate (IPO_4) decomposes upon contact with water, with the only reduction product being I_2 , and no gas is generated.

1-4 In dichloromethane, Br_2 reacts with H_2S in a 1:1 molar ratio, and one of the products is an unstable acid; no element is formed.

1-5 Reactions in non-aqueous solvents can be used to prepare many novel species. Gold(III) fluorosulfate reacts with CO in fluorosulfonic acid,

resulting in a mononuclear complex molecule with Au being sp hybridized; the reaction also releases CO_2 gas. Note the form of ions in non-aqueous solvents.

There are the following statements:

1: The product of the coefficients of all chemical formulas on both sides of the 1-1 equation is 60.

2: The product of the coefficients of all chemical formulas on both sides of the 1-2 equation is 184320.

3: The product of the coefficients of all chemical formulas on both sides of the 1-3 equation is 320.

4: The number of atoms in each molecule on both sides of the 1-4 equation does not exceed 2.

5: The product of the sum of the number of atoms in each type of molecule on both sides of the 1-5 equation is 6912.

The following options are all correct and have the largest number of correct statements:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 5

G. 1,2

H. 1,3

I. 1,4

J. 1,5

K. 2,3

L. 2,4

M. 2,5

N. 3,4

O. 3,5

P. 4,5

Q. 1,2,3

R. 1,2,4

S. 1,2,5

T. 1,3,4

U. 1,3,5

V. 1,4,5

W. 2,3,4

X. 2,3,5

Y. 3,4,5

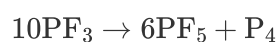
Z. 1,2,3,4

Answer

Correct Answer: S

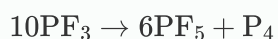
Detailed Explanation

1-1 The disproportionation reaction occurs during the decomposition process. The tetrahedral structure molecule is likely P_4 . The remaining phosphorus element is oxidized to a stable +5 valence state:



CHECKPOINT

1 PTS



The product of the coefficients of all chemical formulas on both sides of the equation is 60, 1 is correct.

1-2 Acetic anhydride $(CH_3CO)_2O$ acts as a dehydrating agent and reaction medium, removing water of hydration and providing an anhydrous environment, promoting the oxidation of nitrate to $Mn(II)$ to $Mn(III)$. Nitrate oxidizes divalent manganese to produce reddish-brown NO_2 . Excess nitric acid is unstable and decomposes under acetic anhydride dehydration. Another component in the mixed gas may be O_2 :



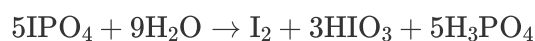
The product of the coefficients of all chemical formulas on both sides of the equation is 184320, 2 is correct.

CHECKPOINT

1 PTS



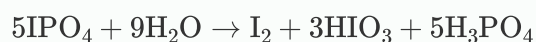
1-3 Trivalent iodine is reduced to zero valence. Then, some iodine undergoes disproportionation to generate stable pentavalent iodate. Balancing the equation:



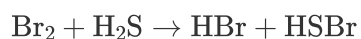
1-3 The product of the coefficients of all chemical formulas on both sides of the equation is 675, not 320, 3 is incorrect.

CHECKPOINT

1 PTS



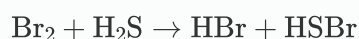
1-4 Under normal circumstances, Br_2 will directly oxidize H_2S to generate elemental sulfur. However, the question indicates a 1:1 reaction, producing an unstable acid and no elemental substance. Therefore, sulfur may be partially oxidized to generate HSBr :



The number of atoms in each molecule on the left and right sides of the equation exceeds 2, 4 is incorrect.

CHECKPOINT

1 PTS



1-5 sp-hybridized gold atoms are two-coordinated. With CO_2 released, it indicates that gold is reduced to the most likely +1 valence state. Considering the difficulty of ion solvation in non-aqueous solvents, a fluorosulfate ligand is needed to directly coordinate and balance the charge, then the other is CO . The oxygen atoms in CO_2 are provided by the fluorosulfate, forming fluorosulfonic anhydride in a non-aqueous solvent:



The product of the sum of the number of atoms in each type of molecule on the left and right sides of the equation is 6912, which is correct.

CHECKPOINT

1 PTS



Therefore, choose S.